Zhouhang Mao
Curriculum Vitæ

Basque Center for
Applied Mathematics
Mazarredo 14, 48009
Bilbao
48009 Bilbao
Spain

Email: first.last AT gmail
DOT com
Homepage: https://
zmao-math.github.io/

Academic Positions

Sep 2026– **Postdoctoral Researcher,** Basque Center for Applied Mathematics, Bilbao, Spain

Sep 2025–Aug 2026 **Postdoctoral Researcher,** Johannes Gutenberg-Universität, Mainz, Germany

Sep 2023–Aug 2025 **Postdoctoral Researcher,** Universiteit van Amsterdam, Amsterdam, Nederland
Mentor: Lenny Taelman

Oct 2021–Aug 2023 **Postdoctoral Researcher,** Institut de Mathématique d'Orsay, Orsay, France
Mentor: Kęstutis Česnavičius

Education

Sep 2018–Oct 2021 **Ph.D. in mathematics,** Sorbonne Université, Paris, France
Thesis: *On derived crystalline cohomology and perfectoid rings as Thom spectra*
Advisor: Matthew Morrow

Sep 2017–Aug 2018 3rd year, École Normale Supérieure, Paris, France

Sep 2016–Aug 2017 **M2 in mathematics,** Université Paris-VII
Mémoire: *Comparisons of different constructions in algebraic K-theory*
Advisor: Matthew Morrow

Sep 2015–Aug 2016 **M1 in mathematics,** École Normale Supérieure, Paris, France

Sep 2012–Aug 2015 **B.Sc. in mathematics,** Fudan University, Shanghai, China

Preprints and publications

Preprint Frobenius–Witt cotangent complex for derived rings, arXiv:2608.04487

Preprint Around Segal conjecture in p-adic geometry, arXiv:2512.17665

Preprint Noncommutative relative de Rham–Witt complex via the norm, arXiv:2410.05998

Accepted Equivariant aspects of de-completing cyclic homology, arXiv:2410.05994, accepted in *Compositio Mathematica*.

Preprint Prismatic logarithm and prismatic Hochschild homology via the norm, arXiv:2409.04400

Published Revisiting derived crystalline cohomology, arXiv:2107.02921, *Bull. Soc. Math. France* 152 (2024), no. 4, 659–784. https://doi.org/10.24033/bsmf.1524zm

Published Perfectoid rings as Thom spectra. *Sel. Math. New Ser.* **29**, 48 (2023). https://doi.org/10.1007/s00029-023-00851-0

Seminar talks

Equivariant aspects of Hochschild homology

21 Nov 2024 Seminar, Universität Regensburg

Revisiting derived crystalline cohomology

09 Nov 2023 Arithmetic and Algebraic Geometry seminar, Universiteit van Amsterdam

10 Feb 2023 Arithmetic Algebraic Geometry, Technical University of Darmstadt

8 Mar 2022 Séminaire d'Homotopie en Géométrie Algébrique, Institut de Mathématiques de Toulouse

9 Nov 2021 Séminaire d'Arithmétique et de Géométrie Algébrique, Université Paris-Saclay

Perfectoid rings as Thom spectra

3 Dec 2020 Seminar of the algebraic topology group, Université Paris Nord

Reviewing activity

Quick Review (1 paper) for Advances in Mathematics, Referee (1 paper) for Compositio Mathematica

Teaching Experience

2025 Spring TA: *Algebraic Geometry 2*

2024 Spring TA: *Modules and categories*

2021 Spring TD L2 *Intégrale de Lebesgue sur* $\mathbf{R}^n$

(Exercise sessions: *Lebesgue integral on* $\mathbf{R}^n$)

2020 Fall TD L2 *Algèbre et Arithmétique*

(Exercise sessions: *Elementary Number Theory and Algebra*)

2020 Spring TD L2 *Algèbre linéaire et bilinéaire IIa*

(Exercise sessions: *Linear and bilinear algebra*)

2019 Fall TD L2 *Séries et Intégrales*

(Exercise sessions: *Series and Integrals*)